Thomas Jiralerspong

Université de Montréal
Mila
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Personal Website
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Google Scholar
LinkedIn
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Education

Université de Montréal
PhD - Computer Science *In progress*
Supervisors: Yoshua Bengio & Guillaume Lajoie
Vanier Canada Graduate Scholarship Scholarship (150 000$)
FRQNT Scholarship (40 000$) (Rank #1 among all applicants in category)
NSERC Canada Graduate Scholarship (17 500$)
Hydro-Québec Excellence Scholarship (10 000$)
Arbour Scholarship (7 500$)

Massachusetts Institute of Technology
Brains, Minds, and Machines Summer Course *2024*

McGill University
B.Sc., Honours Computer Science *2023*
Supervisors: Blake Richards & Doina Precup
GPA: 4.00/4.00
Exchange semester at the **National University of Singapore**
J.W. McConnell Major Entrance Scholarship (9 000$)

Refereed Conferences

Thomas Jiralerspong, Trenton Bricken. ”Cross-Architecture Model Diffing With Crosscoders.” Official Anthropic Blogpost. Under review at ICML 2026.

Luca Scimeca*, **Thomas Jiralerspong***, Berton Earnshaw, Jason Hartford, Yoshua Bengio. ”Learning What Matters: Steering Diffusion via Spectrally Anisotropic Forward Noise.” Under review. 2026.

Eric Elmoznino*, **Thomas Jiralerspong***, Yoshua Bengio, Guillaume Lajoie. “A Complexity-Based Theory of Compositionality.” In *Forty-Second International Conference on Machine Learning (ICML)*. 2025.

Jin Hwa Lee*, **Thomas Jiralerspong***, Lei Yu, Emily Cheng. “Geometric Signatures of Compositionality Across a Language Model’s Lifetime.” In *The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. 2025.

Ezekiel Williams, Avery Ryoo*, **Thomas Jiralerspong***, Matt Perich, Guillaume Lajoie. ”Expressivity of neural networks with random weights and learned biases.” In *The 13th International Conference on Learned Representations (ICLR)*. 2025.

Jean-Pierre Falet, Hae Beom Lee, Nikolay Malkin, Chen Sun, Dragos Secrieru, **Thomas Jiralerspong**, Dinghuai Zhang, Guillaume Lajoie, Yoshua Bengio. “Delta-AI: Local Objectives for Amortized Inference in Sparse Graphical Models” In *Twelfth International Conference on Learning Representations (ICLR)*. 2024.

Chen Sun, Wannan Yang, **Thomas Jiralerspong**, Dane Malenfant, Benjamin Alsbury-Nealy, Yoshua Bengio, Blake Richards. “Contrastive Retrospection: honing in on critical steps for rapid learning and generalization in RL.” In *Thirty-seventh Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2023.

Flemming Kondrup*, **Thomas Jiralerspong***, Elaine Lau, Nathan de Lara, Jacob Shkrob, My Duc Tran, Doina Precup, Sumana Basu. “Towards Safe Mechanical Ventilation Treatment Using Deep Offline Reinforcement Learning.” In *Thirty-seventh AAAI Conference on Artificial Intelligence (AAAI)*. 2023.

Marshall Wang, John Willes, **Thomas Jiralerspong**, Matin Moezzi. “A Comparison of Classical and Deep Reinforcement Learning Methods for HVAC Control.” In *20th IEEE International Conference on Ubiquitous Intelligence and Computing (UIC)*. 2023.

Refereed Workshops

Joachim Schaeffer*, **Thomas Jiralerspong***, Alexander Panfilov*, Roland Zimmerman. “Measuring Control Intervention Awareness Across Frontier LLMs” In *The CAO Workshop at ICLR*. **Best Paper Award**, Oral Presentation. 2026.

Thomas Jiralerspong*, Flemming Kondrup*, Yoshua Bengio. “Noticing the Watcher: LLM Agents Can Infer CoT Monitoring from Blocking Feedback” In *The ICLR Workshop on Agents in the Wild: Safety, Security, and Beyond*. 2026.

Thomas Jiralerspong, Berton Earnshaw, Jason Hartford, Yoshua Bengio, Luca Scimeca. “Shaping Inductive Bias in Diffusion Models through Frequency-Based Noise Control” In *The ICLR Workshop on Deep Generative Models in Machine Learning: Theory, Principle and Efficacy (DeLTa)*. 2025.

Marco Jiralerspong, **Thomas Jiralerspong**, Vedant Shah, Dhanya Sridhar, Gauthier Gidel. “General Causal Imputation via Synthetic Interventions.” In *The Causal Representation Learning Workshop at NeurIPS*. 2024.

Thomas Jiralerspong*, Xiaoyin Chen*, Yash More, Vedant Shah, Yoshua Bengio. “Efficient Causal Graph Discovery Using Large Language Models.” In *How Far Are We From AGI? Workshop at ICLR*. 2024.

Thomas Jiralerspong*, Flemming Kondrup*, Doina Precup, Khimya Khetarpal. “Forecaster: Towards Temporally Abstract Tree-Search Planning from Pixels.” In *Seventh Workshop on Generalization in Planning at NeurIPS*. 2023.

Flemming Kondrup*, **Thomas Jiralerspong***, Elaine Lau, Nathan de Lara, Jacob Shkrob, My Duc Tran, Doina Precup, Sumana Basu. “Deep Conservative Reinforcement Learning for Personalization of Mechanical Ventilation Treatment.” In *The Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*. 2022.

** Equal Contribution

Research Experience

Astra Fellowship - Mentored by Dan Mossing (Anthropic)
Researcher *Apr 2026 - Present*
Project: Characterizing persona space and persona controllability in large language models

Astra Fellowship - Google DeepMind Stream
Researcher *Jan 2026 - Mar 2026*
Project: Study of control intervention awareness in large language models

LawZero
Researcher *Jan 2026 - Mar 2026*
Project: Studied LLM truth recovery in multi-agent environments with strategically deceptive actors

Anthropic - Mentored by Trenton Bricken
Research Fellow *May 2025 - Oct 2025*
Project: Mechanistic interpretability to discover behavioral differences between models

Occam AI
Research Scientist *Jun 2024 - Apr 2025*
Projects: Optimization of interactions between network of LLM agents, automated SQL query generation using LLMs

Waabi - Mentored by Kelvin Wong and Chris Zhang
Deep Learning Research Intern *Jun 2023 – Aug 2023*
Project: Realistic and controllable traffic simulation using a transformer based variational autoencoder

Reasoning and Learning Lab, Mila/McGill University - Supervised by Prof. Doina Precup
Research Intern *Jan 2022 – Aug 2023*
Project: Model-based reinforcement learning with affordance aware tree-search planning directly from pixels

Learning in Neural Circuits Lab, Mila/McGill University - Supervised by Prof. Blake Richards
Research Intern *Sep 2022 – Aug 2023*
Project: Contrastive learning to discover critical states for reinforcement learning in sparse reward environments

Vector Institute for A.I. - Mentored by John Willes and Marshall Wang
Machine Learning Research Intern *Sep 2022 – Dec 2022*
Project: Model-based reinforcement learning for HVAC control

Project X, Machine Learning Research Competition
Co-leader of McGill's Team *Jun 2021 – Feb 2022*
Received the highest score out of 25 submitted papers
Project: Deep offline conservative reinforcement learning for mechanical ventilation treatment

Mentorship

Algoverse AI Research
Principal Investigator *Jun 2026 - Present*
High-level oversight and research direction of 20 AI safety research teams

MARS V (Mentorship for Alignment Research Students) - Meridian Cambridge
Research Mentor *Jun 2026 - Present*
Mentorship of 7 researchers conducting AI safety research, including PhD students (Imperial College London, Polish Academy of Sciences), an ML engineer, and the founder of the Hanoi AI Safety Network

Mila / McGill University
Research interns: Samy Mammeri, Rose Nayoung Kwon, Aly Kassem
Master's student: Élodie Côté-Gauthier

Industry Experience

Amazon Web Services (AWS) – S3 Team
Software Development Engineer Intern *May 2022 – Jul 2022*
Project: JavaScript/Python tool to automate the Incremental Backup recovery system for AWS S3 (stores ∼14 trillion objects)

Square Enix
Software Development Intern *May 2021 – Aug 2021*
Project: Localization system to allow a MOBA game to be translated into over 10 languages

Expedia
Software Development Intern *Jun 2019 – Aug 2019*
Project: React/TypeScript tool to identify which elements of a webpage are broken and conveniently display them to developers

Teaching

Université de Montréal
Teaching Assistant, Representation Learning 2023

McGill A.I. Society
Organizer/Teaching Assistant, Accelerated Intro to ML 2021 – 2023

McGill University
Teaching Assistant, Software Systems 2021 – 2022
Guest Lecturer, Theory of Machine Learning 2022

Honors

Vanier Canada Graduate Scholarship (150000$) 2025

FRQNT Master's Scholarship (40000$) (Rank #1 among all applicants in category) 2024

Arbour Scholarship (7500$)	2024
Hydro-Québec Excellence Scholarship (10000$)	2024
Chosen to attend the 10th Heidelberg Laureate Forum	2023
NSERC Canada Graduate Scholarship (17500$)	2023
University of Montreal Master's Scholarship (5000$)	2023
McGill Mobility Bursary for Exchanges (6000$)	2022
Winner of UofT AI's Project X competition (25000$)	2022
J.W. McConnell Major Entrance Scholarship (9000$)	2020 – 2022
CIBPA Foundation Bursary (1000$, 2500$, 1000$)	2021, 2022, 2023
Marianopolis College Valedictorian	2020
Governor General of Canada's Academic Medal	2020

Invited Talks

Canadian Undergraduate Conference on AI (CUCAI)	2022
University of Toronto AI Conference	2022
McGill AI Society Learnathon	2022

Professional Activities

Mila

Chairman of Lab Representatives	2023 – Present
Chairman of Social Committee	2023 – Present
Executive Member of Recruitment Committee	2023 – Present

McGill AI Society

Senior Advisor	2023 – Present
Technical Project Manager	2021 – 2023

Montreal AI & Neuroscience Conference

Organizer – Introduction to deep learning with PyTorch workshop — 2022

McGill NeuroTech

Machine Learning Developer — 2021 – 2022

McGill Robotics

Software Developer — 2020 – 2021

Languages

Native: English, French
Advanced: Italian, Spanish
Beginner: Mandarin, Japanese

Skills

Programming Languages: Python, Java, JavaScript, R, C, C++, C#, OCaml, SQL, HTML, CSS

Machine Learning Libraries: PyTorch, TensorFlow, Keras, Pandas, NumPy, Matplotlib

Other: LaTeX, Slurm, Jupyter Notebooks, Perforce, GitHub, Jira, Unity

Press

Anthropic. Mar 13, 2026. A “Diff” Tool for AI: Finding Behavioral Differences in New Models. Official Anthropic blog post.

SciLogs. Nina Beier. Jan 24, 2024. What Do Food and Research Have in Common? More Than You Might Think.

The McGill Tribune. Mikaela Shadick. March 15, 2022. Six McGill undergrads win UofT international artificial intelligence competition.

McGill Reporter. Richard Deschamps. March 1, 2022. Undergrad team uses machine learning to create a better hospital ventilator.

Advanced Coursework

Université de Montréal
Representation Learning
Reinforcement Learning & Optimal Control
Scaling Laws
Causal Inference & Machine Learning
Probabilistic Graphical Models

McGill University
Reinforcement Learning
Brain Inspired Artificial Intelligence
Honours Math for Machine Learning
Probabilistic Programming
Network Science

National University of Singapore
Quantum Computing
Information Theory